

Conditional finiteness for

$$x^3y^2 = z^4 + 1$$

A self-contained proof assuming the *abc* conjecture

The equation has only finitely many integer solutions *if the abc conjecture holds*. All steps below are elementary once that conjecture is assumed. This proves the implication; it does not establish finiteness independently of the conjecture.

1. Precise statement and proof

For a positive integer n , its *radical* $\text{rad}(n)$ is the product of its distinct prime divisors, with $\text{rad}(1) = 1$. In particular, $\text{rad}(n) \leq n$ and $\text{rad}(mn) \leq \text{rad}(m)\text{rad}(n)$ for positive integers m, n . These inequalities follow directly by counting each prime at most once.

The abc conjecture. *For every $\varepsilon > 0$ there is a constant $K_\varepsilon \geq 1$ such that all pairwise coprime positive integers a, b, c satisfying $a + b = c$ obey*

$$c \leq K_\varepsilon \text{rad}(abc)^{1+\varepsilon}.$$

Theorem 1. *Assume the abc conjecture, and let $K = K_{1/6}$. Every integer solution of $x^3y^2 = z^4 + 1$ satisfies*

$$1 \leq x \leq K^{8/3}, \quad 1 \leq |y| \leq K^4, \quad |z| \leq K^2.$$

Consequently the equation has only finitely many integer solutions.

Proof. The right-hand side is positive, so $y \neq 0$ and $x \geq 1$. If $z = 0$, then $x^3y^2 = 1$, giving exactly $(x, y, z) = (1, 1, 0), (1, -1, 0)$; both satisfy the claimed bounds because $K \geq 1$.

Now put $t = |z| \geq 1$ and $N = t^4 + 1 = x^3y^2$. Each prime divisor p of N divides x or y , and hence its exponent in N is at least two. Unique prime factorization therefore gives

$$\text{rad}(N)^2 \mid N, \quad \text{and thus} \quad \text{rad}(N) \leq N^{1/2}. \quad (1)$$

The integers $1, t^4, N$ are pairwise coprime: any common divisor of t^4 and N divides $N - t^4 = 1$. Also $\text{rad}(t^4) = \text{rad}(t) \leq t \leq N^{1/4}$, so

$$\text{rad}(1 \cdot t^4 \cdot N) \leq \text{rad}(t^4) \text{rad}(N) \leq N^{1/4} N^{1/2} = N^{3/4}. \quad (2)$$

Applying *abc* with $\varepsilon = 1/6$ to $1 + t^4 = N$ yields

$$N \leq K \text{rad}(t^4 N)^{7/6} \leq K N^{7/8}.$$

Since $N > 0$, division gives $N^{1/8} \leq K$, hence $N \leq K^8$. Finally $t^4 < N$, $x^3 \leq N$, and $y^2 \leq N$ give the stated bounds. Only finitely many integer triples lie in these bounded ranges. $\square$

2. Intuition: why the exponents force a bound

The useful feature of x^3y^2 is that it has no prime factor occurring only once. Such a positive integer is called *powerful*. Its size can grow much faster than its radical: a prime power p^e contributes p^e to the integer but only p to the radical. Because every exponent here is at least two, the radical uses at most half of the size exponent, as expressed by (1).

The natural additive identity is

$$\underbrace{1}_{\text{radical } 1} + \underbrace{t^4}_{\text{radical at most } N^{1/4}} = \underbrace{N}_{\text{radical at most } N^{1/2}}.$$

Thus the product of all distinct primes in this coprime triple is at most $N^{3/4}$. The conjecture says that the sum N cannot substantially exceed that product raised to a power arbitrarily close to one. The gap

$$\frac{1}{4} + \frac{1}{2} = \frac{3}{4} < 1$$

is precisely what makes a finiteness proof possible.

More explicitly, any fixed $0 < \varepsilon < 1/3$ gives

$$N \leq K_\varepsilon N^{\frac{3}{4}(1+\varepsilon)}, \quad N^{\frac{1-3\varepsilon}{4}} \leq K_\varepsilon, \quad N \leq K_\varepsilon^{\frac{4}{1-3\varepsilon}}.$$

The exponent on the middle left-hand side is positive. Since K_ε is independent of the solution, N cannot grow without bound. Choosing $\varepsilon = 1/6$ simply makes this exponent $1/8$.

3. Scope of the conclusion

No coprimality assumption on x and y is needed. The required coprimality comes from the consecutive integers t^4 and $t^4 + 1$. Negative x and $y = 0$ are impossible by positivity; both signs of y and z are included through absolute values.

The two solutions $(1, \pm 1, 0)$ are explicit. The argument does not assert that these are the only solutions: it proves that the total solution set is finite under *abc*. The stated form of the conjecture supplies no numerical value for $K_{1/6}$, so the bound above is not a numerical search cutoff and does not by itself produce a complete list.

The proof uses no external results beyond the stated conjecture and elementary unique prime factorization. No internet search or computational search is needed.